

REMARKS TO MY PAPER “ON THE DISTRIBUTION OF ADDITIVE AND THE MEAN VALUES OF MULTIPLICATIVE ARITHMETIC FUNCTIONS”

By

G. HALÁSZ (Budapest)

Let $\mathfrak{P}$ be an arbitrary set of primes with

$$E(x) \stackrel{\text{def.}}{=} \sum_{\substack{p \leq x \\ p \in \mathfrak{P}}} \frac{1}{p} \rightarrow \infty \quad (x \rightarrow \infty)$$

and $g(n)$ be the completely additive number-theoretic function (i.e. $g(mn) = g(m) + g(n)$) defined by

$$g(p) = \begin{cases} 1 & \text{if } p \in \mathfrak{P}^1 \\ 0 & \text{otherwise,} \end{cases}$$

i.e. $g(n)$ denotes the number of prime divisors of n from the arbitrary set $\mathfrak{P}$, counted with multiplicity. In [1], referred to in the title, we proved that

$$N(m, x) \stackrel{\text{def.}}{=} \sum_{\substack{p \leq x \\ g(n)=m}} 1 \begin{cases} \sim x \frac{E^m(x)}{m!} e^{-E(x)} & \text{if } \frac{m}{E(x)} \sim 1, \\ \gg x \frac{E^m(x)}{m!} e^{-E(x)} & \text{if } 1 - c \leq \frac{m}{E(x)} \leq 1 + c,^2 \\ \ll x \frac{E^m(x)}{m!} e^{-E(x)} & \text{if } \delta \leq \frac{m}{E(x)} \leq 2 - \delta. \end{cases}$$

uniformly for the integer m .

Here we add the following result.

¹ Throughout p stands for primes.

² $c, c_1, \dots$ denote positive constants depending possibly on δ if δ is relevant. $a \ll b$ means $\frac{a}{b} \leq c_i$.

THEOREM.

$$N(m, x) = \sum_{\substack{n \leq x \\ g(n)=m}} 1 \begin{cases} \ll x \frac{E^m(x)}{m!} e^{-E(x)} & \text{whenever } 0 \leq m \leq (2 - \delta)E(x) \\ \gg x \frac{E^m(x)}{m!} e^{-E(x)} & \text{whenever } \delta E(x) \leq m_0 \leq (2 - \delta)E(x), \text{ either} \\ & \text{for } m = m_0 \text{ or } m = m_0 + 1. \end{cases}$$

The first statement is an extension of the previous upper bound to a larger range, but note the difference between the second statement and the previous lower bound: we do not know whether the lower bound is valid for *each* m in $\delta E(x) \leq m \leq (2 - \delta)E(x)$ outside the interval $(1 - c)E(x) \leq m \leq (1 + c)E(x)$.³ (For the background of the whole problem see [1].)

Our method is simple and "almost" elementary (compare with that of [2]). We remark, however, that for the asymptotic relation, especially with the sharp remainder term of [1], the more complicated analytic methods of [1] are needed (see the remark after formula (6) of the present paper), even if they could have been somewhat simplified, had our intentions in [1] been to prove that result only.

PROOF. Consider

$$(1) \quad \sum_{\substack{n=1 \\ g(n)=m}}^{\infty} \frac{1}{n^{\sigma}} = \int_1^{\infty} \frac{dN(m, u)}{u^{\sigma}} = \sigma \int_1^{\infty} \frac{N(m, u)}{u^{\sigma+1}} du \quad (\sigma > 1)$$

by partial integration. Following A. SELBERG [2], as we did in [1], we express this in terms of

$$F(z, \sigma) = \sum_{n=1}^{\infty} \frac{z^{g(n)}}{n^{\sigma}} \quad (z = re^{i\theta}, r \leq 2 - \delta)$$

by integration with respect to z :

$$(2) \quad \sum_{\substack{n=1 \\ g(n)=m}}^{\infty} \frac{1}{n^{\sigma}} = \sum_{n=1}^{\infty} \frac{1}{n^{\sigma}} \cdot \frac{1}{2\pi i} \int_{|z|=r} z^{g(n)-m-1} dz = \frac{1}{2\pi i} \int_{|z|=r} \frac{F(z, \sigma)}{z^{m+1}} dz.$$

Here $z^{g(n)}$ is the completely multiplicative function (i.e. $f(mn) = f(m)f(n)$) defined by

$$f(p) = \begin{cases} z & \text{if } p \in \mathfrak{P} \\ 1 & \text{otherwise,} \end{cases}$$

so that

$$F(z, \sigma) = \prod_{p \in \mathfrak{P}} \frac{1}{1 - \frac{z}{p^{\sigma}}} \prod_{p \notin \mathfrak{P}} \frac{1}{1 - \frac{1}{p^{\sigma}}} = e^{\sum_p' \sum_{k=1}^{\infty} \frac{z^k}{kp^{k\sigma}} + \sum_p'' \sum_{k=1}^{\infty} \frac{1}{kp^{k\sigma}}}.$$

³ This will be deduced from our present results by A. Sárközy in a subsequent paper.

Here and in what follows Σ' and Σ'' mean summation over primes $p \in \mathfrak{P}$ and $p \notin \mathfrak{P}$, respectively. The product and all series converge absolutely for $\sigma > 1$, $\frac{|z|}{p} \leq \frac{|z|}{2} < 1$.

In order to investigate F , consider

$$\frac{F(z, \sigma)}{F(r, \sigma)} = e^{\sum_p' \sum_{k=1}^{\infty} \frac{z^k - r^k}{kp^{k\sigma}}}$$

We can disregard terms with $k \geq 2$, since

$$\begin{aligned} \sum_p' \sum_{k=2}^{\infty} \frac{|z^k - r^k|}{kp^{k\sigma}} &\leq |z - r| \sum_p' \sum_{k=2}^{\infty} \frac{(2 - \delta)^k}{p^k} \leq \\ &\leq r |\vartheta| \sum_p' \frac{(2 - \delta)^2}{p^2 \left(1 - \frac{2 - \delta}{p}\right)} \leq c_1 r |\vartheta|. \end{aligned}$$

Defining y by $\sigma - 1 = \frac{1}{\log y}$, we can also drop terms with $p > y$, $k = 1$ and replace σ by 1 for the remaining $p \leq y$, $k = 1$, making the same error as before, owing to the elementary estimates

$$\sum_{p > y} \frac{1}{p^\sigma} \leq c_2 \quad (\text{see (6) of [1]})$$

$$\sum_{p \leq y} \left(\frac{1}{p} - \frac{1}{p^\sigma} \right) \leq c_3 \quad (\text{see (8) of [1]}).$$

Hence

$$\begin{aligned} \frac{F(z, \sigma)}{F(r, \sigma)} &= e^{\sum_{p \leq y}' \frac{z-r}{p} + O(r|\vartheta|)} = e^{(z-r)E(y)} (1 + O(r|\vartheta|)) = \\ &= e^{(z-r)E(y)} + O(e^{r(\cos \vartheta - 1)E(y)} r |\vartheta|). \end{aligned}$$

Putting this into (2), the first term on the right yields exactly the m th Taylor coefficient of $e^{(z-r)E(y)}$ in z , i.e. $\frac{E^m(y)}{m!} e^{-rE(y)}$. The contribution of the remainder is less than

$$\frac{c_4}{r^m} \int_{-\pi}^{\pi} e^{-c_5 \vartheta^2 r E(y)} r |\vartheta| d\vartheta \leq \frac{c_4}{r^m E(y)} \int_{-\infty}^{\infty} e^{-c_5 u^2} |u| du = \frac{c_6}{r^m E(y)}.$$

In order to minimize this error compared to the previous main term we set $r = \frac{m+1}{E(y)}$

which is legitimate, provided $\frac{m+1}{E(y)} \leq 2 - \delta$ and Stirling's formula shows that in

this case it becomes in fact inferior by a $\sqrt{E(y)}$ to the main term. We have thus obtained

$$\sum_{\substack{n=1 \\ g(n)=m}}^{\infty} \frac{1}{n^{\sigma}} = F(r, \sigma) \frac{E^m(y)}{m!} e^{-rE(y)} \left(1 + O\left(\frac{1}{\sqrt{E(y)}} \right) \right).$$

Here, estimating as we have done before,

$$\begin{aligned} F(r, \sigma) &= e^{\sum_p' \sum_{k=1}^{\infty} \frac{p^k-1}{kp^{k\sigma}} + \sum_p' \sum_{k=1}^{\infty} \frac{1}{kp^{k\sigma}}} = e^{\sum_p' \frac{r-1}{p^{\sigma}} + O(1)} F(1, \sigma) = \\ &= e^{\sum_{p \leq y}' \frac{r-1}{p} + O(1)} \frac{1}{\sigma-1} = e^{(r-1)E(y) + O(1)} \log y \end{aligned}$$

by the first order pole of $\zeta(\sigma) = F(1, \sigma)$, the Riemann zeta function. Combining our results with (1),

$$(3) \quad c_7 \frac{E^m(y)}{m!} e^{-E(y)} \log y \leq \int_1^{\infty} \frac{N(m, u)}{u^{\sigma+1}} du \leq c_8 \frac{E^m(y)}{m!} e^{-E(y)} \log y$$

$$\left(m+1 \leq (2-\delta)E(y), \sigma-1 = \frac{1}{\log y} \right).$$

For the upper estimation choose $\sigma-1 = \frac{1}{\log x}$, i.e. $y = x$. If $u \leq x$, then $u^{\sigma-1} \leq x^{\frac{1}{\log x}} = e$ and we get further

$$(4) \quad \int_1^x \frac{N(m, u)}{u^{\sigma}} du \leq e \int_1^{\infty} \frac{N(m, u)}{u^{\sigma+1}} du \leq c_9 \frac{E^m(x)}{m!} e^{-E(x)} \log x.$$

We make another use of the number-theoretic properties. Introducing the Mangoldt symbol

$$\Lambda(n) = \begin{cases} \log p & \text{if } n = p^k \\ 0 & \text{otherwise,} \end{cases}$$

$$(5) \quad \sum_{\substack{n \leq u \\ g(n)=m}} \log n = \sum_{\substack{n \leq u \\ g(n)=m}} \sum_{d|n} \Lambda(d) = \sum_{d \leq u} \Lambda(d) \sum_{\substack{k \leq u/d \\ g(kd)=m}} 1 = \sum_{d \leq u} \Lambda(d) N\left(m - g(d), \frac{u}{d}\right).$$

The contribution of $d = p^k$ with $k \geq 2$ is trivially

$$O\left(\sum_{\substack{p^k \\ k \geq 2}} \log p \cdot \frac{u}{p^k} \right) = O(u).$$

For $d = p$ either $g(d) = 1$ ($p \in \mathfrak{P}$) or $g(d) = 0$ ($p \notin \mathfrak{P}$) so that

$$\begin{aligned} \sum_{\substack{n \leq u \\ g(n)=m}} \log n &= \sum'_{p \leq u} \log p N\left(m-1, \frac{u}{p}\right) + \sum''_{p \leq u} \log p N\left(m, \frac{u}{p}\right) + O(u) \leq \\ &\leq \Sigma \log p \left[N\left(m-1, \frac{u}{p}\right) + N\left(m, \frac{u}{p}\right) \right] + O(u). \end{aligned}$$

(We have omitted $p \leq u$ as $N\left(m, \frac{u}{p}\right) = 0$ for $p > u$ anyway. Terms with $m-1$ are to be cancelled when $m = 0$.) For $u \geq x$ the left hand side is greater than

$$\sum_{\substack{n \leq x \\ g(n)=m}} \log n \geq [N(m, x) - N(m, \sqrt{x})] \log \sqrt{x} \geq \frac{1}{2} [N(m, x) - \sqrt{x}] \log x$$

and integration gives

$$\begin{aligned} x \frac{1}{2} [N(m, x) - \sqrt{x}] \log x &\leq \int_x^{2x} \sum_{\substack{n \leq u \\ g(n)=m}} \log n \, du \leq \\ &\leq \Sigma \log p \int_1^{2x/p} \left[N\left(m-1, \frac{u}{p}\right) + N\left(m, \frac{u}{p}\right) \right] du + O(x^2) = \\ &= \sum_p p \log p \int_1^{2x/p} [N(m-1, u) + N(m, u)] du + O(x^2) = \\ &= \int_1^x [N(m-1, u) + N(m, u)] \sum_{p \leq 2x/u} p \log p \, du + O(x^2). \end{aligned}$$

Here

$$\sum_{p \leq 2x/u} p \log p \leq \frac{2x}{u} \sum_{p \leq 2x/u} \log p \leq c_{10} \frac{x^2}{u^2}$$

giving rise to integrals estimated in (4):

$$x[N(m, x) - \sqrt{x}] \log x \leq c_{11} x^2 \left(\frac{E^{m-1}(x)}{(m-1)!} + \frac{E^m(x)}{m!} \right) e^{-E(x)} \log x + O(x^2),$$

and since $m \ll E(x)$,

$$N(m, x) \leq c_{12} \left(x \frac{E^m(x)}{m!} e^{-E(x)} + \frac{x}{\log x} \right)$$

under the condition $\frac{m+1}{E(x)} \leq 2 - \delta$ which, δ being arbitrary, is the same as that

in our theorem. It remains to show that $\frac{x}{\log x}$ is superfluous or, as

$$e^{-E(x)} \geq e^{\sum_{p \leq x} \frac{1}{p}} = e^{-\log \log x + O(1)} \geq \frac{c_{13}}{\log x},$$

that

$$\frac{E^m(x)}{m!} \geq c_{14}.$$

The quantity increases for $m \leq E(x)$, decreases for $E(x) \leq m \leq 2E(x)$ and attains, therefore, its minimal value at $m = 0$ or $m = 2E(x)$. The first case is trivial and in the second

$$\frac{[E(x)]^{2E(x)}}{((2E(x))!)} \geq c_{15} \frac{[E(x)]^{2E(x)}}{\left(\frac{2E(x)}{e}\right)^{2E(x)} \sqrt{E(x)}} = \frac{c_{15}}{\sqrt{E(x)}} \left(\frac{e}{2}\right)^{2E(x)} \rightarrow \infty,$$

thus completing the proof of the upper bound.

The lower estimation is again based on the identity (5). The left hand side is smaller than $N(m, x) \log x$ for $u \leq x$, while we may drop the non-negative terms with $d = p^k$, $k \geq 2$ on the right. Integrating for $1 \leq u \leq x$ we get the same way as before

$$(6) \quad xN(m, x) \log x \geq \int_1^x N(m-1, u) \sum'_{p \leq x/u} p \log p \, du + \int_1^x N(m, u) \sum''_{p \leq x/u} p \log p \, du.$$

In want of any good estimation for the inner sums, in the corresponding step for the upper bound we replaced Σ' and Σ'' roughly by Σ (and this is the reason why this method cannot supply asymptotic formulae). Here we can do the same by writing down this last inequality for $m+1$ as well, keeping only the second integral for m and the first for $m+1$ and adding the two inequalities:

$$x[N(m, x) + N(m+1, x)] \log x \geq \int_1^x N(m, u) \left[\sum''_{p \leq x/u} p \log p + \sum'_{p \leq x/u} p \log p \right] du.$$

For the complete sum $\Sigma = \Sigma'' + \Sigma'$ we have already the elementary inequality

$$\sum_{p \leq x/u} p \log p \geq \frac{x}{2u} \sum_{x/2u \leq p \leq x/u} \log p \geq c_{16} \frac{x^2}{u^2}$$

for $x/u \geq 2$, i.e. $u \leq x/2$ so that

$$(7) \quad [N(m, x) + N(m+1, x)] \log x \geq x \int_1^{x/2} \frac{N(m, u)}{u^2} du.$$

In order to be able to apply (3), we first have to eliminate the infinite part of its integral. For this purpose we make use of the upper bound already proven in our theorem with $\delta/2$ in place of δ . If $\frac{m}{E\left(\frac{x}{2}\right)} \leq 2 - \delta/2$, then it is applicable for $u \leq x/2$

giving

$$\int_{x/2}^{\infty} \frac{N(m, u)}{u^{\sigma+1}} du \ll \frac{1}{m!} \int_{x/2}^{\infty} \frac{E^m(u) e^{-E(u)}}{u^{\sigma}} du.$$

Generally ($\beta > \alpha$, $m \leq 2E(\alpha)$)

$$\begin{aligned} \frac{E^m(\beta) e^{-E(\beta)}}{E^m(\alpha) e^{-E(\alpha)}} &= e^{E(\alpha) - E(\beta) + m \log \frac{E(\beta)}{E(\alpha)}} \leq e^{E(\alpha) - E(\beta) + m \left[\frac{E(\beta)}{E(\alpha)} - 1 \right]} \leq \\ (8) \quad &\leq e^{E(\beta) - E(\alpha)} \leq e^{\sum_{\alpha < p \leq \beta} \frac{1}{p}} \leq c_{17} e^{\log \frac{\log \beta}{\log \alpha}} = c_{17} \frac{\log \beta}{\log \alpha}. \end{aligned}$$

Choosing now $\alpha = x/2$, $\beta = u$

$$\begin{aligned} \frac{1}{m!} \int_{x/2}^{\infty} \frac{E^m(u) e^{-E(u)}}{u^{\sigma}} du &\leq \frac{c_{17}}{\log \frac{x}{2}} \frac{E^m\left(\frac{x}{2}\right) e^{-E\left(\frac{x}{2}\right)}}{m!} \int_{x/2}^{\infty} \frac{\log u}{u^{\sigma}} du \leq \\ &\leq \frac{c_{18}}{\log x} \frac{E^m\left(\frac{x}{2}\right) e^{-E\left(\frac{x}{2}\right)}}{m!} \frac{2^{\sigma-1}}{x^{\sigma-1}} \left(\frac{1}{(\sigma-1)^2} + \frac{\log x}{\sigma-1} \right). \end{aligned}$$

In case this is smaller than, say, the half of the left of (3), i.e.

$$c_{19} \frac{E^m\left(\frac{x}{2}\right) e^{-E\left(\frac{x}{2}\right)}}{x^{\sigma-1} \log x} \left(\frac{1}{(\sigma-1)^2} + \frac{\log x}{\sigma-1} \right) \leq E^m(y) e^{-E(y)} \log y$$

with a sufficiently large c_{19} , $\int_{x/2}^{\infty}$ can in fact be dropped in (3) with c_7 replaced by its half. Let $\log y = \frac{1}{\sigma-1} = \frac{1}{K} \log x$ ($K > 1$). Using (8) with $\alpha = y$, $\beta = x/2$ the condition becomes

$$c_{20} \frac{\log x}{\log y} e^{-\frac{\log x}{\log y}} \left(\frac{\log y}{\log x} + 1 \right) \leq 1, \quad c_{20} K e^{-K} \left(\frac{1}{K} + 1 \right) \leq 1$$

and indeed, this holds for sufficiently large fixed $K = c_{21}$. Hence we have

$$\int_1^{x/2} \frac{N(m, u)}{u^2} du \geq \int_1^{x/2} \frac{N(m, u)}{u^{\sigma+1}} du \geq \frac{c_7}{2} \frac{E^m(y)e^{-E(y)}}{m!} \log y \geq c_{21} \frac{E^m(x)e^{-E(x)}}{m!} \log x,$$

using again (8), provided $\frac{m}{E(y)} \leq 2 - \delta/2$. Since, however, $\frac{m}{E(x)} \leq 2 - \delta$ as

assumed in our theorem and $E(x) - E(y) \leq \sum_{y < p \leq x} \frac{1}{p} \leq c_{22}$, this will be satisfied for large enough x . Putting our last result into (7)

$$N(m, x) + N(m+1, x) \geq x \frac{E^m(x)e^{-E(x)}}{m!}$$

and either

$$N(m, x) \geq x \frac{E^m(x)e^{-E(x)}}{m!}$$

or

$$N(m+1, x) \geq x \frac{E^m(x)e^{-E(x)}}{m!} \geq x \frac{E^{m+1}(x)e^{-E(x)}}{(m+1)!}$$

(recall $\frac{m}{E(x)} \geq \delta$) and the proof is completed

(Received 25 January 1972)

MTA MATEMATIKAI KUTATÓ INTÉZET
BUDAPEST, V., REÁLTANODA U. 13–15

References

- [1] G. HALÁSZ, On the distribution of additive and the mean values of multiplicative arithmetic functions, *Studia Sci. Math. Hung.*, **6** (1971), pp. 211–233.
- [3] H. DELANGE, Un théorème sur les fonctions arithmétiques multiplicatives et ses applications, *Annal. Sci. Éc. Norm. Sup.*, **3** ser **78** (1961), pp. 1–29.
- [3] A. SELBERG, Note on a paper by L. G. Sathe, *J. Indian Math. Soc.*, **18** (1954), pp. 83–87.